QUANTARA AI — FULL PROJECT EXPLAINED

Comprehensive Institutional Research Suite: Machine Learning, Deep Learning, & Market Regime Economics

Date: October 30, 2026 | **Universe:** 8 Liquid NSE Blue-Chips | **Research Scope:** 2017–2026

MANDATORY RESEARCH NOTICE: Quantara AI is an experimental quantitative research platform. All findings represent historical backtest simulations. Past performance does not guarantee future results. **Live trading is strictly disabled; no broker APIs or order-routing mechanisms exist.**

1. Executive Overview & Research Objective

Quantara AI was created to rigorously evaluate whether predictive artificial intelligence can systematically forecast medium-term equity trends in the Indian stock market after factoring in authentic transaction costs and market friction. The project analyzes 8 highly liquid National Stock Exchange (NSE) securities: **BHARTIARTL, HDFCBANK, ICICIBANK, INFY, ITC, RELIANCE, SBIN, and TCS** spanning nearly a decade of authentic daily price history (2017–2026, 19,380+ observations). The primary prediction target evaluates the **20-trading-session forward return (~1 calendar month)**, executing orders at Open t+1 based on Close t signals.

2. Chronological Research Journey (Phases 1 to 26)

• **Phases 1–4: Data Hygiene & Leakage Prevention:** Audited 19,776 raw records from zero-synthetic source data. Engineered 20 technical indicators. Established strict 20-day embargoes between chronological splits to eliminate forward-return leakage. Verified scalers fit only on historical training rows.

• **Phases 13–15B: The Phase 13 Machine Learning Baseline:** Standardized on 15 trend features (group_1_trend). Established the authoritative Phase 13 Logistic Regression baseline: 52.63% test balanced accuracy (47.28% standard accuracy, 46.32% macro F1). Confirmed that unconstrained long-only ML trading lost money (-17.46%) due to cumulative trading costs and bear market drawdowns.

• **Phases 16–19: Institutional Cost Models & Dividend Recovery:** Formally integrated India's statutory delivery fee schedule (0.3937% round-trip covering STT, exchange turnover, SEBI charges, stamp duty, GST, and brokerage). Extracted 124 authentic corporate dividend events to credit cash on ex-dates.

• **Phases 20–23: Breakthrough — Dynamic Cash Allocation (Rules D, E, F):** Investigated market-regime filtering. Discovered that capital preservation is driven by dynamically holding cash during hostile regimes rather than aggressive trading. Rule F (dynamic regime allocation) protected capital by holding up to 100% cash in market downturns. Rules D, E, and F were mathematically audited and frozen.

• **Phases 24–25: Deep Learning Implementation & Forensic Audit:** Implemented 2-layer LSTM (56,162 params) and 2-layer GRU (42,658 params) using 60-day sequences. Phase 25 forensic audit identified two critical defects in Phase 24: (1) An invalid majority-baseline log loss of 21.9177 caused by passing hard labels [0, 1] into scikit-learn, and (2) Walk-forward folds 1–3 were evaluated on static models without retraining (in-sample training memorization).

• **Phase 26: Definitive Corrective Run & True Walk-Forward Retraining:** Corrected the baseline probability bug (valid log loss = 0.7794). Executed true walk-forward retraining across 4 distinct expanding folds (4,647 out-of-sample observations). Proved that recurrent deep learning (LSTM: 49.92%, GRU: 48.68%) did NOT outperform simple Logistic Regression (50.07%).

3. Definitive Out-of-Sample Performance Comparison

Below is the verified performance across **4,647 genuine out-of-sample observations** (2023–2026) evaluated under true walk-forward retraining with 20-day block bootstrap confidence intervals:

Model Architecture	Balanced Acc	95% Block Bootstrap CI	Macro F1	Log Loss	Brier Score	Paired Diff vs LR
Corrected Majority Baseline	50.00%	[50.00%, 50.00%]	35.75%	0.6977	0.2521	N/A

Logistic Regression (Retrained)	**50.07%**	[46.19%, 54.25%]	**49.91%**	**0.6942**	**0.2505**	Baseline
LSTM Classifier (Fresh Retrained)	49.92%	[44.96%, 55.11%]	46.89%	0.6983	0.2524	-0.15% (CI spans 0)
GRU Classifier (Fresh Retrained)	48.68%	[45.73%, 51.44%]	39.37%	0.6992	0.2528	-1.39% (CI spans 0)

4. Core Scientific Takeaways

• **Complexity Does Not Yield Alpha:** Moving from a 16-parameter linear model to a 56,162-parameter recurrent neural network did not improve predictive performance. Financial markets exhibit high noise and non-stationary distribution shifts that easily cause complex models to overfit spurious past sequences.

• **Mode Collapse Vulnerability:** Freshly retrained LSTM and GRU models suffered from persistent mode collapse toward the historical training class ('UP'), producing UP recall of 79%–93% while DOWN recall collapsed to 6%–21%. When market conditions turned sharply negative (2025–2026 downtrend), deep networks failed to adapt.

• **Risk Management Dominates Prediction:** Phases 20–23 confirmed that risk-adjusted capital preservation stems from dynamic cash management (Rule F regime filters) rather than trying to predict low-conviction directional price moves.

• **True Walk-Forward Evaluation is Crucial:** Testing models on data they were already trained on created an illusion of 55%–60% accuracy in Phase 24. Only true chronological retraining revealed actual out-of-sample performance (~50%).

5. Planned Roadmap: Phase 27 Hypothetical Rs. 1,00,000 Backtest

The next planned research phase (Phase 27) will calculate the actual hypothetical rupee profit or loss from investing Rs. 1,00,000 starting capital using an **unfiltered 3-class target** (DOWN: < -2%, HOLD: [-2%, +2%], UP: > +2%). Because future outcomes cannot be filtered out in live trading, Phase 27 will simulate long-only execution (Predict UP: enter equity; Predict HOLD/DOWN: hold cash) with equal max 12.5% allocation across the 8 stocks, full statutory friction (0.3937% delivery fees + 0.05% slippage), dividend credits, and comparisons against Buy-and-Hold and Idle Cash benchmarks.